

# Caleb O'Neal

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## EDUCATION

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| <b>Bachelor of Science in Computer Engineering</b><br>University of Virginia           | Expected May 2028<br>GPA: 3.94/4.00 |
| <b>Associate of Science in Computer Science</b><br>Northern Virginia Community College | June 2025<br>GPA: 3.84/4.00         |

## WORK EXPERIENCE

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| <b>Intern</b><br>Virginia Tech Qualcomm Thinkabit Lab                                                                                                                                                                                                                                                                                                                                                                 | August 2024 – May 2025<br>Falls Church, VA |
| <ul style="list-style-type: none"><li>Built an interactive web-based simulation of a three-phase brushless DC motor to teach robotics students motor-control fundamentals using React and Rust</li><li>Assisted in lab tasks like cleaning, organizing, developing teaching materials, leading school groups, and moving the lab between buildings</li></ul>                                                          |                                            |
| <b>Teaching Assistant</b><br>Arlington Career Center                                                                                                                                                                                                                                                                                                                                                                  | January 2025 – May 2025<br>Arlington, VA   |
| <ul style="list-style-type: none"><li>Taught a 15 student college statistics course, teaching statistical analysis and real-world data interpretation</li><li>Evaluated assignments and research posters, critiquing students’ analysis of relationship in real-world datasets</li></ul>                                                                                                                              |                                            |
| <b>High School Intern</b><br>Department of Defense                                                                                                                                                                                                                                                                                                                                                                    | June 2024 – August 2024<br>Virginia        |
| <ul style="list-style-type: none"><li>Redesigned a department website to guide agency management to enterprise architecture data visualization tools</li><li>Migrated the team’s Git repositories between Git hosts, enabling new CI/CD features and better DX for the team</li><li>Collaborated with the team’s Scrum Master to follow Agile principles including regular code reviews and sprint planning</li></ul> |                                            |

## PROJECTS

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| <b>PyCon US 2024 Poster Presentation</b> , NASA App Development Challenge                                                                                                                                                                                                                                                                                                                                                     | October 2023 – May 2024 |
| <ul style="list-style-type: none"><li>Developed a 3D interactive simulation of lunar surface exploration for a potential landing site for NASA Artemis III mission</li><li>Selected to present our simulation to software professionals at the PyCon US 2024 poster presentation session</li></ul>                                                                                                                            |                         |
| <b>Martian Crater Data Analysis</b>                                                                                                                                                                                                                                                                                                                                                                                           |                         |
| <ul style="list-style-type: none"><li>Analyzed a dataset of 350,000 Martian craters and found a statistical relationship between crater eccentricity and its latitude on Mars using a pearson correlation test</li><li>Used SAS to perform statistical analysis and visualize geographic trends in the dataset</li><li>Presented findings for peer review and to an audience at Northern Virginia Community College</li></ul> |                         |
| <b>Captain, Programming Lead</b> , Robotics Team                                                                                                                                                                                                                                                                                                                                                                              | August 2022 – May 2025  |
| <ul style="list-style-type: none"><li>Managed a team of 50 with an annual budget of \$50,000 to design, build, wire, program, and test a robot designed for the 2024 and 2025 FIRST Robotics Competitions</li><li>Redesigned the team management structure following significant growth in members</li><li>Led a team of 10 programmers using GitHub’s pull request and issue systems for code review</li></ul>               |                         |

## SKILLS

- Programming:** Python, C, C++, Java, JavaScript, SAS, Rust, Zig
- Tools:** Git, CI/CD, Linux, KiCad, QEMU